

Quiz 06 v1**Name:** _____*Essential Skills – Must show work for full credit.***1.** Find the vertex of $f(x) = -3x^2 + 12x - 9$.**2.** Describe the behavior of
 $f(x) = (x - 2)^2(x + 4)$ at each zero.**3.** Write a polynomial with roots -1 and 5 .**4.** Determine the degree, leading coefficient, and
end behavior of $g(x) = -4x^5 + 2x^3 - 1$.

5. Expand and evaluate $\sum_{i=1}^5 i^2$.	6. Rewrite $4 + 8 + 12 + 16$ using sigma notation.
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Quiz 06 v2**Name:** _____*Essential Skills – Must show work for full credit.*

1. Convert $f(x) = x^2 - 8x + 7$ into vertex form.

2. Determine the minimum possible degree if roots are -3 (mult 2) and 2 .

3. Construct a degree 3 polynomial with roots $0, 1, -4$.

4. For $f(x) = 2x^4 - 6x^2$, identify symmetry and end behavior.

5. Evaluate $\sum_{i=0}^4 (i + 1)$.

6. Explain what the bounds mean in $\sum_{k=2}^6 k$.

Quiz 06 v3**Name:** _____*Essential Skills – Must show work for full credit.*

1. Identify the axis of symmetry of
 $f(x) = 4x^2 - 16x + 3$.

2. Sketch the end behavior of
 $h(x) = -2x^6(x - 1)$.

3. A quadratic has x-intercepts at 2 and -3. Write one possible equation.

4. For $f(x) = (x + 1)^2(x - 4)$, state multiplicities and crossing behavior.

5. Simplify $\sum_{k=1}^4 3k$.

6. Explain why constants factor out of summations.

Quiz 06 v4**Name:** _____*Essential Skills – Must show work for full credit.*

1. Graph $f(x) = (x - 2)^2 - 3$ and identify the vertex.

2. Determine whether $g(x) = -5x^4 + 3x$ has both ends up, down, or opposite.

3. Write a polynomial that crosses at -2 and touches at 3.

4. For $p(x) = x^5 - 5x^3 + 4x$, determine degree and symmetry.

5. Evaluate $\sum_{i=1}^3 (2i + 1)$.	6. In your own words, explain what a summation represents.
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Quiz 06 v5**Name:** _____*Essential Skills – Must show work for full credit.*

1. Complete the square: $x^2 + 6x + 4$.

2. Identify multiplicity of $f(x) = (x - 5)^3$.

3. Find a polynomial with roots -1, 0, and 4 and leading coefficient 2.

4. Determine the maximum number of turning points of a degree 7 polynomial.

5. Separate $\sum_{i=1}^4 (i^2 + 2i)$.

6. Evaluate $\sum_{k=1}^5 k$.

Quiz 06 v6**Name:** _____*Essential Skills – Must show work for full credit.***1.** Determine the range of $f(x) = -x^2 + 4x - 2$.**2.** Analyze $f(x) = -3(x - 2)^2(x + 1)$ for roots and end behavior.**3.** Write a degree 4 polynomial with roots 1 (mult 2) and -2 (mult 2).**4.** Explain why a polynomial of degree n has at most $n - 1$ turning points.

5. Evaluate $\sum_{i=0}^3 (i^2 - 1)$.	6. Explain the meaning of the bounds in $\sum_{i=3}^7 f(i)$.
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Answer Key — Quiz 06 Versions

Quiz 06 v1

- 1. Vertex: (2, 3)
- 2. Zeros at $x = 2$ (mult 2), $x = -4$ (mult 1)
- 3. Example: $(x + 1)(x - 5)$
- 4. Degree 5; leading coefficient negative; end behavior up left, down right
- 5. 55
- 6. $4(1) + 4(2) + 4(3) + 4(4)$

Quiz 06 v2

- 1. Vertex form: $(x - 4)^2 - 9$
- 2. Minimum degree 3
- 3. Example: $x(x - 1)(x + 4)$
- 4. Even degree; both ends up
- 5. 10
- 6. Summation adds values from lower bound to upper bound

Quiz 06 v3

- 1. Axis of symmetry: $x = 2$
- 2. Even degree; negative leading coefficient; both ends down
- 3. Example: $(x - 2)(x + 3)$
- 4. Root -1 (mult 2), root 4 (mult 1)
- 5. 30
- 6. Constants factor out of summations

Quiz 06 v4

- 1. Vertex: (2, -3)
- 2. Opposite directions (odd degree, negative leading coefficient)
- 3. Example: $(x + 2)(x - 3)^2$
- 4. Degree 5; odd function
- 5. 15
- 6. Summation represents repeated addition

Quiz 06 v5

- 1. $(x + 3)^2 - 5$
- 2. Multiplicity 3
- 3. Example: $2x(x - 1)(x + 4)$
- 4. 6 turning points maximum
- 5. Separate into two summations
- 6. 15

Quiz 06 v6

- 1. Range: $y \leq 2$
- 2. Roots 2 (mult 2), -1 (mult 1); odd degree, negative leading coefficient
- 3. Example: $(x - 1)^2(x + 2)^2$
- 4. Derivative has degree $n - 1$
- 5. 10
- 6. Bounds indicate starting and ending index

Quiz 06 – Holistic Grading Rubric

Essential Skills (Front Page – 75%)

Grade	Description
A	Shows correct mathematical structure and reasoning. Work is organized and complete. Minor arithmetic errors at most.
B	Correct approach and structure shown. Several computational errors or small reasoning gaps.
C	Partial understanding. Correct idea attempted but structure incomplete or inconsistent.
D	Minimal correct reasoning shown. Major gaps in structure or misunderstanding of key concepts.
Not Yet	No meaningful evidence of understanding.

Above & Beyond (Back Page – 25%)

Grade	Description
A	Correct structure and reasoning. No computational errors. Clear understanding of extended concept.
B	Correct structure. Minor computational errors.
C	Correct idea attempted but incomplete or inconsistent reasoning.
D	Limited understanding shown.
Not Yet	No meaningful attempt.

Teacher Notes

- Must show work for full credit.
- Emphasis is on mathematical structure and reasoning.
- Front page evaluates core objectives.
- Back page evaluates extended reasoning and precision.